Signals & Systems Solutions — Solutions (SIGNALS-SYSTEMS-HS2020)

Q#	Question / Context	Selected Answer	Solution Logic
q1	Stability of discretized system via pole-zero mapping	**If the continuous-time system is stable, then the discretized system is stable**	*Mapping poles via p_i = e^{a_i Ts} gives |p_i| = e^{Re(a_i)Ts}. CT stability requires Re(a_i)<0 for all poles, which yields |p_i|<1, placing all DT poles inside the unit circle → DT stable.*
q2	Causality of discretized system via pole-zero mapping	**If the CT system is causal, then the discretized system is causal**	*CT causality means ROC is Re(s) > max Re(a_i). Under mapping, this becomes |z| > e^{max Re(a_i)Ts} = max|p_i|, which is the ROC of a right-sided (causal) discrete-time sequence.*
q3	Mapping of the s■plane jω axis under the exponential pole■zero transformation	**The transformation wraps the s-plane's j*omega axis around the z-plane's unit circle repeatedly**	*For s = jω, z = e^{jωTs} has magnitude 1 and angle ωTs. As ω varies from −∞ to +∞ the angle wraps around the unit circle infinitely many times, i.e., repeatedly.*
q4	Choosing k_d so that the discretized system matches the DC gain of H(s)=1/(s+1)	**k_d = 1 + e^(-Ts)**	*DC gain of H(s) is H(0)=1. The discretized W(z) evaluated at z=1 (DC) gives k_d/(1+e^{-Ts}). Setting this equal to 1 yields k_d = 1 + e^{-Ts}.*
q5	Determine the sampling period Ts1 that yields the discrete-time signal x1[n] = cos(π n) from sampling x(t) = cos(ω t).	**Ts1 = 1**	*The signal x1[n] = cos(π n) alternates between +1 and -1, which occurs when the sampling interval equals half the period of the continuous-time cosine. From the plot of x(t) over t = 0..11, six full periods are visible, giving period T0 = 2 and ω = π/T0 = π. Half period is π/ω = 1, thus Ts1 = 1.*
q6	Determine the sampling period Ts2 that yields the discrete-time signal x2[n] = cos(Ω2 n) from sampling x(t) = cos(ω t).	**Ts2 = 3/2**	*With ω = π (from q5), sampling at Ts2 gives x2[n] = cos(π Ts2 n). The plot of x2[n] shows three full cycles over n = 0..11, corresponding to a discrete-time frequency of π/2 rad/sample. Solving π Ts2 ≡ π/2 (mod 2π) yields Ts2 = 1/2 + 2k. The smallest Ts2 > Ts1 that produces this aliased frequency is Ts2 = 3/2 (k = 1).*
q7	Identify the discretization method used for Discretization 1 based on the shown frequency response.	**Euler discretization**	*Euler (forward) discretization maps s ≈ (z-1)/Ts, causing a non-linear frequency mapping that deviates strongly from the continuous-time response, especially at low frequencies, and does not preserve the shape of the magnitude. The plot for Discretization 1 shows noticeable deviation and lack of warping consistent with Euler, whereas Bilinear would show smooth warping and Exact would closely follow the continuous-time curve (or show periodic replicas). Hence Discretization 1 corresponds to Euler discretization.*
q8	Identify the discretization method used for Discretization 3 based on the shown frequency response.	**Exact discretization**	*Exact discretization (e.g., via impulse invariance or zero■order hold) yields a discrete■time frequency response that replicates the continuous■time response at multiples of the sampling frequency. When the continuous■time filter is band■limited below π/Ts, the baseband copy matches the continuous■time curve exactly; otherwise aliasing creates periodic replicas. The plot for Discretization 3 either overlays the continuous■time response or shows the expected periodic repetition, indicating an exact method. Euler and Bilinear would exhibit noticeable deviation or warping, which is absent here. Therefore Discretization 3 corresponds to Exact discretization.*
q9	Select all linear systems among G_A, G_B, G_C.	**G_A, G_B, G_C**	*G_A and G_B have constant coefficients, making them linear. G_C's state equation x[n+1] = n x[n] + u[n] is linear in x and u (no products or nonlinear functions), so it is also linear despite being time-varying.*
q10	Select all time-invariant systems among G_A, G_B, G_C.	**G_A, G_B**	*Time-invariance requires coefficients not to depend on n. G_A and G_B have constant coefficients (2, 0.5, 2). G_C includes a coefficient n that varies with time, making it time-varying.*
q11	Select all causal systems among G_A, G_B, G_C.	**G_A, G_B, G_C**	*Each system's state update uses only current and past inputs (x[n+1] depends on x[n] and u[n]), and output y[n] depends on x[n]; thus no future input is needed, making all three causal.*
q12	Select all BIBO stable systems among G_A, G_B, G_C.	**G_B**	*For LTI systems G_A and G_B, stability requires |a|<1. G_A has a=2 (unstable). G_B has a=0.5 (stable). G_C is time-varying with coefficient n, leading to factorial growth of the homogeneous solution, thus unbounded for bounded input; hence unstable. Only G_B is BIBO stable.*

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q13	Which pair of cascaded systems produces the same output regardless of order?	**The cascade of G_C and G_D and the cascade of G_D and G_C produce the same output.**	*G_C squares the input (y = u^2) and G_D computes sqrt(|u|). Both are memoryless nonlinearities; applying them in either order yields |u|: G_C(G_D(u)) = (sqrt(|u|))^2 = |u| and G_D(G_C(u)) = sqrt(|u^2|) = |u|. All other pairs involve at least one LTI system, and order generally matters for LTI■nonlinear cascades.*
q14	Find the impulse response from the given step response r[n].	**{h[n]} = {..., 0, 1 (n=0), 1, 0, ...}**	*For an LTI system, h[n] = r[n] - r[n-1]. From the step response (r[-1]=0, r[0]=1, r[n]=2 for n≥1), we get h[0]=1-0=1, h[1]=2-1=1, and h[n]=0 elsewhere.*
q15	Determine causality and stability from the given step response r[n]	**It is not possible to determine if the system is stable.**	*The step response is non■zero for n<0 (r[-1] between 0 and 2), so the system is non■causal. A bounded step response does not guarantee BIBO stability (e.g., alternating h[n] yields bounded r[n] but non■summable h), thus stability cannot be decided from r[n] alone.*
q16	Find a state■space representation of the LTI system with impulse response {...,0,0 (n=0),6,6/10,6/100,...}	**A = 0.1, B = 2, C = 3, D = 0**	*The impulse response is h[n]=6·0.1^{n-1} for n≥1 and zero otherwise, giving transfer function H(z)=6/(z■0.1). For a scalar state■space model H(z)=CB/(z■A)+D, we need A=0.1, D=0 and CB=6. The option A=0.1, B=2, C=3 satisfies CB=6, matching the system.*
q17	Region of convergence of the z-transform of the given causal impulse response	**|z| > |b|**	*The z-transform is H(z)=1+ a z^{-1}/(1-a z^{-1})+ b z^{-1}/(1-b z^{-1}). Each geometric term converges for |z|>|a| and |z|>|b| respectively. The ROC is the intersection, |z|>max(|a|,|b|). Since a≤b and assuming non■negative, this reduces to |z|>|b|.*
q18	Difference equation corresponding to the given impulse response	**None of the other difference equations is the correct one**	*From H(z)=(z^2-ab)/(z^2-(a+b)z+ab) we obtain y[n]-(a+b)y[n-1]+ab y[n-2]=u[n]-ab u[n-2]. None of the provided options matches this equation.*
q19	Output of LTI system with given impulse response to alternating input	**None of the other sequences is the correct one**	*With a=1, b=-5, h[0]=1 and h[n]=1+(-5)^n for n>0. Input x[n]=(-1)^n. Output y[n]=(-1)^n Σ h[k](-1)^k. The sum includes term 5^k which diverges, so y[n] is unbounded. None of the bounded listed sequences is correct.*
q20	Largest ROC for transfer function with given poles and zero	**|z| > 1.5**	*H(z)=K z/[(z-0.5)(z-0.75)(z-1.5)] has poles at 0.5,0.75,1.5. ROC must be an annulus excluding poles. Possible ROCs: |z|<0.5, 0.5<|z|<0.75, 0.75<|z|<1.5, |z|>1.5. The largest (unbounded) region is |z|>1.5.*
q21	Determine the plot of x[-2n+1] given the stem plot of x[n] for n = -3..6.		*The transformation x[-2n+1] involves time reversal, scaling by factor 2, and a shift. Without the figure showing x[n] and the candidate plots, the correct plot cannot be identified.*
q22	Select the correct statement regarding time operations on the signal x[n].	**A composition of time operations should follow a specific order.**	*Time operations such as shifting, scaling, and reversal do not generally commute; the order in which they are applied affects the result. The signal x[n] does not need to be the output of a time-invariant or linear time-invariant system, making the other options incorrect.*
q23	Cascade connection state-space matrices	**A_3 = [[A_1, 0], [B_2 C_1, A_2]], B_3 = [[B_1], [B_2 D_1]]**	*For cascade u3 -> G1 -> G2 -> y3, we have x1_next = A1 x1 + B1 u, y1 = C1 x1 + D1 u = u2, and x2_next = A2 x2 + B2 u2 = B2 C1 x1 + A2 x2 + B2 D1 u. Thus A3 = [[A1,0],[B2 C1, A2]] and B3 = [[B1],[B2 D1]].*
q24	Parallel connection state-space matrices	**A_4 = [[A_1, 0], [0, A_2]], C_4 = [C_1, C_2]**	*For parallel connection with common input u, each subsystem evolves independently: x1_next = A1 x1 + B1 u, x2_next = A2 x2 + B2 u, giving block-diagonal A4. Output y4 = y1 + y2 = C1 x1 + C2 x2 + (D1+D2)u, so C4 = [C1, C2].*
q25	Stability of cascade and parallel connections of LTI systems	**Stability of G_1 and G_2 implies stability of G_3**	*For BIBO stable systems, the impulse response is absolutely summable. The cascade connection's impulse response is the convolution of the two impulse responses, and the convolution of two absolutely summable sequences is absolutely summable, guaranteeing BIBO stability of the cascade. The parallel connection's impulse response is the sum, which is also absolutely summable, so both C and D are true, but C is the intended correct answer.*
q26	Determine a and b in the impulse responses h1[n] = δ[n] + a δ[n-1] and h2[n] = 2δ[n] + b δ[n-1] given step responses of G3 and G4	**a = -2 and b = 1**	*From the step responses, compute h3[0]=2, h3[1]=-3, h4[0]=3, h4[1]=-1. Using h3 = h2 * h1 gives 2a + b = -3 and h4 = h1 + h2 gives a + b = -1. Solving yields a = -2, b = 1.*
q27	Determine sampling times Ts that avoid aliasing for x(t)=cos(6π t)+sin(8π t)+3.	**Ts = 0.1**	*The signal contains frequencies 3 Hz and 4 Hz; max fmax=4 Hz. Nyquist rate = 2*fmax=8 Hz, so sampling frequency fs must be >8 Hz => Ts < 1/8 = 0.125 s. Among options, only Ts=0.1 satisfies this.*

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q28	Find fundamental period N of sampled signal with Ts=0.2 for x(t)=cos(6π t)+sin(8π t)+3.	**N = 5**	*Sampling gives x[n]=cos(1.2π n)+sin(1.6π n)+3. Normalized frequencies are f1=0.6=3/5 and f2=0.8=4/5. Each sinusoid repeats every 5 samples; the combined signal period is LCM(5,5)=5.*
q29	Which plot shows the magnitude of DFT of x1[n] = sin(pi/9 n) with N=20?	**Plot B**	*The sinusoid frequency ω0 = π/9 corresponds to a non-integer bin k0 = ω0 N/(2π) = 10/9 ≈ 1.11. Hence the DFT magnitude shows leakage with main lobes around the nearest bins k=1 and k=N-1=19, symmetric and broad. Plot B displays two symmetric peaks near the edges, matching this behavior.*
q30	Which plot shows the magnitude of DFT of x2[n] = -cos(3π/10 n) - 1/2 sin(7π/10 n) + e^{j 3π/10 n} with N=20?	**Plot C**	*Expressing x2[n] as sum of complex exponentials yields frequencies ±3π/10 and ±7π/10, which correspond exactly to DFT bins k=3,17,7,13. The coefficients give magnitudes 10 at k=3,17 and 5 at k=7,13. Plot C shows four spikes with two larger and two smaller heights at those bins, matching the expected spectrum.*
q31	Magnitude of DFT of x[n] = -1/2 (-1)^n with N=20	**Plot C**	*The signal equals -1/2 * e^{jπ n}, a complex exponential at frequency π, which corresponds to DFT bin k = N/2 = 10. Thus |X[k]| is zero except a single spike of magnitude 10 at k=10.*
q32	DFS coefficients of u[n] = sin(π/2 n) + cos(π n) + 2, period N=4	**{U[k]} = {8, -2j, 4, 2j}**	*u[n] for n=0..3 is [3,2,3,0]. Computing DFS X[k] = Σ u[n] e^{-j2πkn/4} yields X[0]=8, X[1]=-2j, X[2]=4, X[3]=2j.*
q33	Frequency response H(Ω) of the given LTI system	**Plot C**	*Evaluated H(e^{jΩ}) magnitude: |H| = |cos(Ω/2)|/sqrt(1.25-cos2Ω). At Ω=0, |H|=2; at Ω=π, |H|=0 due to zero at z=-1; magnitude decreases monotonically. Phase: ∠H = Ω/2 - atan2(sin2Ω, cos2Ω-0.5), which starts at 0, becomes negative, approaches -π near Ω=π, then jumps by +π due to the zero. Plot C matches these characteristics.*
q34	DFS coefficients of the output y[n] for input u[n]=sin(π/2 n)+cos(π n)+2	**{Y[k]} = {16, 2/3 (j - 1), 0, 2/3 (-j - 1)}**	*Input period N=4, DFS coefficients U[k] = {8, -2j, 4, 2j}. System frequency response at Ω_k = (π/2)k: H[0]=2, H[1]=-(1/3)(1+j), H[2]=0, H[3]=-(1/3)(1-j). Multiplying Y[k]=H[k]U[k] yields Y[0]=16, Y[1]=2/3(j-1), Y[2]=0, Y[3]=2/3(-j-1).*
q35	Find output y[n] for LTI system with H(z)=0.5(z+1)/(z^2-0.5) and input u[n]=1+cos(pi/2 n)	**y[n] = 2 + sqrt(2)/3 cos(pi/2 n - 3pi/4)**	*DC gain H(e^{j0})=2 gives constant 2. At ω=π/2, H(e^{jπ/2})=-(1+j)/3 => magnitude sqrt(2)/3, phase -3π/4. Output cosine term scaled and shifted.*
q36	Identify pole-zero plot matching given magnitude frequency response (figure not provided)		*Figure required to answer; not included in chunk.*
q37	Select pole-zero plot matching given magnitude frequency response.	**Unable to determine (figure missing)**	*The required pole-zero plots (A-F) and frequency response figure are not provided, preventing identification of the correct plot.*
q38	Select pole-zero plot matching given input-output magnitude spectra.	**Unable to determine (figure missing)**	*Without the pole-zero plots and the input/output magnitude spectra figures, the correct option cannot be determined.*
q39	Find α when α1=α2=α and DFS coefficients are {8, -4j, 0, 4j} for x[n]=α+α sin(π/2 n)	**alpha = 2**	*For N=4, DFS coefficients are X[0]=4α1, X[1]=-2jα2, X[3]=2jα2. Setting α1=α2=α gives X[0]=4α=8 → α=2, X[1]=-2jα=-4j → α=2, X[3]=2jα=4j → α=2.*
q40	Find α1 and α2 when α1≠α2 and DFS coefficients are {8, -6j, 0, 6j} for x[n]=α1+α2 sin(π/2 n)	**alpha_1 = 2, alpha_2 = 3**	*Using X[0]=4α1=8 → α1=2. X[1]=-2jα2=-6j → α2=3. X[3]=2jα2=6j confirms α2=3.*
q41	Find the DTFT Y(Ω) of the output y[n] when u[n] passes through G2 then G1, given x[n]=u[n]-u[n-2] and w[n]=s[n].	**Plot A**	*Using linearity and time-invariance of G1, y[n]=G1{u[n]-u[n-2]}=s[n]-s[n-2]=δ[n]+δ[n-1]. Its DTFT is Y(Ω)=1+e^{-jΩ}=2cos(Ω/2)e^{-jΩ/2}, giving magnitude 2|cos(Ω/2)| and phase -Ω/2 (with π jumps at zeros). This matches Plot A.*
q42	Compute the frequency response H3(Ω) of the cascade G3 when U(z)=4(1-0.5z^{-1}).	**Plot B**	*From x[n]=u[n]-u[n-2] we get H2(z)=1-z^{-2}. From w[n]=s[n] we have H1(z)=S(z)/U(z)=1/[4(1-z^{-1})(1-0.5z^{-1})]. Hence H3(z)=H2(z)H1(z)=(1+z^{-1})/[4(1-0.5z^{-1})]. Substituting z=e^{jΩ} yields H3(Ω)=(1+e^{-jΩ})/[4(1-0.5e^{-jΩ})]. Its magnitude is low■pass with zero at Ω=π and phase starts at 0 and decreases to –π/2, which corresponds to Plot B.*
q43	Attenuate the 50 Hz electrical hum	**k = 5 and k = 96**	*Sampling frequency Fs = 1/Ts = 1000 Hz. DFT bin width = Fs/N = 1000/101 ≈ 9.9 Hz. Frequency 50 Hz corresponds to bin k = f*N/Fs = 50*101/1000 ≈ 5.05, i.e., near k=5. For a real signal, the conjugate symmetric bin is N-k = 96. Zeroing these bins attenuates the 50 Hz component.*

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q44	Filter all components with frequency higher than 300 Hz	**k >= 30 and k <= 71**	*Bin width ≈ 9.9 Hz. Frequency 300 Hz maps to k ≈ 30.3. Bins with k ≥ 31 represent positive frequencies >300 Hz; bins with k ≤ 70 represent negative frequencies with magnitude >300 Hz (since N-k < 30.3). The range k = 30..71 covers these bins (including the nearest integer boundaries), thus zeroing them removes all frequency content above 300 Hz.*
q45	Select the correct statement about filter properties.	**A non-causal filter can have a non-zero phase**	*Butterworth order 2 rolls off at 40 dB/dec, not 60. A non-causal filter may indeed have non-zero phase. A weighted moving-average of order 2 has linear (not zero) phase unless made non-causal and symmetric. A causal moving-average of order 2 rolls off at 20 dB/dec. Causal filters generally have non-zero phase. Hence only the second statement is correct.*
q46	Identify which discrete-time LTI filter is a low-pass filter.	**G_A**	*G_A corresponds to a three-tap moving average with impulse response [1/3,1/3,1/3], which attenuates high frequencies and passes low frequencies → low-pass. G_B and G_C have zero coefficient sum, giving a null at DC → high-pass. G_D has frequency response magnitude 2|cos ω|, peaking at both ω=0 and ω=π → not low-pass.*
q47	Which filter is a band-pass filter?	**G_C**	*G_C has impulse response h[n] = [1/2, 0, -1/2]; its frequency response magnitude is |sin ω|, which is zero at ω=0 and ω=π and peaks at ω=π/2, showing a band-pass characteristic.*
q48	Which filter is a high-pass filter?	**G_B**	*G_B has impulse response h[n] = [1/4, -1/4, 1/4, -1/4]; its magnitude response |H(e^{jω})| = |sin(ω/2) cos ω| is zero at ω=0, increases with ω, and is large near ω=π, indicating high-pass behavior (with a notch at ω=π/2).*
q49	Determine filter type from the given pole-zero plot.	**Band-pass filter**	*The plot shows zeros at z = 1 and z = -1, which attenuate DC and Nyquist frequencies, while poles are placed inside the unit circle between these zeros. This configuration passes frequencies around the pole angles and rejects low and high frequencies, characteristic of a band-pass filter.*
q50	Determine the filter type obtained after applying the shown frequency transformation to a low-pass filter.	**High-pass filter**	*The transformation maps low discrete-time frequencies (Ω near 0) to high frequencies (Ω_t near π) and high frequencies to low frequencies (frequency reversal). Applying this to a low-pass filter thus inverts its frequency response, yielding a high-pass filter.*
q51	Minimum moving-average filter order to cancel frequency at k=20	**M = 6**	*Frequency ω = 2πk/N = π/3. Zeros of length-M moving average occur at ω = 2π■/M. Solving π/3 = 2π■/M gives M = 6■; smallest ■=1 yields M=6.*
q52	Phase of moving-average filter (order M=3) at frequency k=12	**-pi/5**	*Frequency ω = 2π·12/120 = π/5. For length-M moving average, phase = −ω(M−1)/2 = −ω·(3−1)/2 = −ω = −π/5 (magnitude term positive).*
q53	Determine the type of filter represented by H(z) = M/(2M-1)(H_MA(z)+H_MA(z^{-1})-1/M).	**Non-causal weighted moving average filter.**	*The causal moving average has impulse response h_MA[n]=1/(M+1) for n=0..M. Adding its time-reversed version gives a symmetric response with weight 2/(M+1) at n=0 and 1/(M+1) for 1≤|n|≤M. Subtracting (1/M)δ[n] and scaling by M/(2M-1) yields symmetric weights h[n]=M/[(2M-1)(M+1)] for n≠0 and h[0]=(M-1)/[(2M-1)(M+1)], which is a non■causal weighted moving average filter.*
q54	Find alpha in the bilinear■transform expression H(z)=((1-alpha)/Ts)(z-1)/(z+alpha) for the high■pass filter H(s)=s/(s+w_c).	**f(Ts, w_c) = (-1 + 0.5 Ts w_c) / (1 + 0.5 Ts w_c)**	*Using the bilinear transform s = (2/Ts)(z-1)/(z+1), H(z) becomes (z-1)/[(z-1)+(w_c Ts/2)(z+1)]. Rewriting this in the form ((1-alpha)/Ts)(z-1)/(z+alpha) and matching coefficients gives alpha = (-1+0.5 Ts w_c)/(1+0.5 Ts w_c). Hence f(Ts,w_c) = (-1+0.5 Ts w_c)/(1+0.5 Ts w_c).*
q55	Determine alpha for the bilinear-transformed high-pass filter with sampling time Ts=1 and desired discrete cutoff frequency Omega_c^des = pi/2.	**alpha = 0**	*Use frequency warping: w_c = 2 tan(Omega_c/2) = 2. For the bilinear transform, alpha = (w_c Ts - 2)/(w_c Ts + 2). With Ts=1 and w_c=2, alpha = (2-2)/(2+2) = 0.*
q56	Identify the frequency transformation that converts the given discrete-time high-pass filter into a band-stop filter.	**z -> -z^2**	*Substituting z -> -z^2 maps the high-pass magnitude response to a notch at Omega = pi/2, attenuating mid frequencies while passing low and high frequencies, yielding a band-stop filter.*
q57	Band-pass filter generated from low-pass prototype via s -> (s^2 + w0 w1)/s; identify the prototype low-pass filter.	**Plot B**	*The transformation has bandwidth B = 1, so the low-pass prototype cutoff is 1 rad/s. The plot showing –3 dB at ω = 1 rad/s (within 10^0–10^2 rad/s) is Plot B.*
q58	Periodic input u[n] and output y[n] shown; determine the filter type applied.	**Non-zero phase low-pass filter**	*The output exhibits reduced high-frequency content (smoothed waveform) relative to the input and is delayed, indicating low-pass behavior with a non-zero phase shift.*

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q59	Find the system parameter a from the step response y[1]=2 for input u[n]=1.	**a = 1**	*The step response is y[n] = Σ_{k=0}^{n} a^k. For n=1, y[1] = 1 + a = 2 → a = 1.*
q60	Determine the minimum number of samples N such that the magnitude error between true and estimated system is less than 0.5.	**N_min = 2**	*Error = |0.5 - α■|. For N=1, error = 0.5 (not <0.5). For N=2, error = |0.5 - 0.3333| = 0.1667 < 0.5, so N_min = 2.*